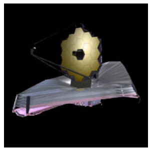

2072 - See Through Supernovae: Nebular Spectroscopy of Exploding White Dwarfs

Cycle: 1, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Saurabh W. Jha (PI)	**Rutgers the State University of New Jersey**
Lindsey Kwok (CoI)	Rutgers the State University of New Jersey
Ms. Yssavo Camacho-Neves (CoI)	Rutgers the State University of New Jersey
Mr. Kyle Dettman (CoI)	Rutgers the State University of New Jersey
Dr. John P. Hughes (CoI)	Rutgers the State University of New Jersey
Prof. Ryan Foley (CoI)	University of California - Santa Cruz
Dr. Armin Rest (CoI)	Space Telescope Science Institute
Dr. Curtis McCully (CoI)	Las Cumbres Observatory Global Telescope Network
Dr. Carles Badenes (CoI)	University of Pittsburgh
Dr. Tea Temim (CoI)	Princeton University
Dr. Ori Dosovitz Fox (CoI)	Space Telescope Science Institute
Dr. Louis-Gregory Strolger (CoI)	Space Telescope Science Institute
Dr. Joel Pearson Johansson (CoI) (ESA Member)	Stockholm University
Dr. Tamas Szalai (CoI) (ESA Member)	University of Szeged

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI LRS + NIRSPEC PRISM WDSN-2				
	1	LRS-Slit	MIRI Low Resolution Spectroscopy	(10) SN2021AEFX
	2	NIR PRISM	NIRSpec Fixed Slit Spectroscopy	(10) SN2021AEFX
	3	LRS-Slit	MIRI Low Resolution Spectroscopy	(10) SN2021AEFX
	4	NIR PRISM	NIRSpec Fixed Slit Spectroscopy	(10) SN2021AEFX
MIRI LRS + NIRSPEC PRISM WDSN-3				

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
	5	LRS-Slit	MIRI Low Resolution Spectroscopy	(12) SN2022PUL
	6	NIR PRISM	NIRSpec Fixed Slit Spectroscopy	(12) SN2022PUL
	7	LRS-Slit	MIRI Low Resolution Spectroscopy	(12) SN2022PUL
	8	NIR PRISM	NIRSpec Fixed Slit Spectroscopy	(12) SN2022PUL
MIRI MRS + NIRSPEC G395M/G235M WDSN-1				
	9	MRS	MIRI Medium Resolution Spectroscopy	(11) SN2022AAIQ
	10	NIR G395M+G235M	NIRSpec Fixed Slit Spectroscopy	(11) SN2022AAIQ
	11	MRS	MIRI Medium Resolution Spectroscopy	(11) SN2022AAIQ
	12	NIR G395M+G235M	NIRSpec Fixed Slit Spectroscopy	(11) SN2022AAIQ

ABSTRACT

Type Ia supernovae (SN Ia) have enormous importance to cosmology and astrophysics, but we still lack a detailed understanding of their progenitor systems and explosion mechanisms. At late times, in the nebular phase, the ejecta become optically thin, allowing us to "see through" the supernova and directly probe its composition, density, temperature, and kinematic structure. Nebular phase observations in the near-infrared and mid-infrared provide unique and powerful constraints on models, including the density-dependent nucleosynthesis of intermediate mass elements, radioactive iron-group elements, and stable iron-group elements. Here we propose to build a legacy, reference sample of JWST near- and mid-infrared nebular spectra of white dwarf supernovae with non-disruptive target-of-opportunity observations. In addition to normal SN Ia, we will also observe the diversity of thermonuclear supernovae, including extreme and peculiar objects. Our data will map progenitor and explosion scenarios to supernova outcomes, explaining different classes and narrowing the allowed model space for normal SN Ia. All data obtained will be made public immediately, with no exclusive access period.

OBSERVING DESCRIPTION

We will observe 9 target-of-opportunity white dwarf supernovae (WDSN) with 3 objects per cycle for 3 cycles. Each supernova will be observed in two epochs to measure the spectral evolution and disentangle radioactive and stable elements. All epochs consist of mid-infarared (MIRI) and near-infrared (NIRSpec) observations. For one object per cycle, we will use the MIRI MRS and NIRSpec G235M+G395 to cover a wide wavelength range from 1.7 to 28 microns in both epochs at relatively high spectral resolution. The other two objects per cycle will use MIRI LRS and the NIRSpec prism to obtain spectroscopic coverage from 0.6 to 13 microns at low resolution (R ~ 100). The low-resolution observations can target fainter, more distant WDSN, while the higher-resolution data will target, rarer, more nearby objects.

During the program we will trigger on WDSN from different classes, including normal SN Ia, extreme SN Ia (91bg/86G-like and 91T/99aa-lick objects), and peculiar objects (SN Iax, Ia-CSM, super-Chandra SN Ia, etc.) ensuring that the breadth of the thermonuclear supernova zoo is adequately represented in our sample, while still mainting a focus on SN Ia. We will trigger on objects that are well-observed in ground-based data near maximum light, and nearby enough (with limiting distance depending on the specific type of object) to get adequate S/N to achieve our science goals.

Our nebular-phase observations will be scheduled approximately +150 and +250 days past maximum light for each supernova. Thus, these will be non-disruptive TOOs that can be scheduled months in advance. We will work with the WDSN community to identify the best targets, and all data collected in this program will be made publicly available immediately with no exclusive access period.

Because we are obtaining spectroscopy of supernovae that change with time, the MIRI and NIRSpec observations need to be nearly contemporaneous for each epoch. Thus we have linked those observations as non-interruptible. This also leads to decreased slew overhead so that the charged time will match the allocated time when we trigger.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	SN2021AEFX	RA: 04 19 53.4000 (64.9725000d) Dec: -54 56 53.09 (-54.94808d) Equinox: J2000		
Comments: *Category=Star* *Description=[Type Ia supernovae]* *Extended=NO*				
(11)	SN2022AAIQ	RA: 14 26 32.0290 (216.6334542d) Dec: +56 35 3.13 (56.58420d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae, Type Ia supernovae]* *Extended=NO*				
(12)	SN2022PUL	RA: 12 26 48.8460 (186.7035250d) Dec: +08 26 55.32 (8.44870d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae, Type Ia supernovae]* *Extended=NO*				

Generic Targets

#	Name	Criteria	Description
(4)	C2-LONGTERM-TOO-WDSN-4-MEDRES	nearby white dwarf supernova	
Comments: LONGTERM TOO for Cycle 2			
(5)	C2-LONGTERM-TOO-WDSN-5-LOWRES	nearby white dwarf supernova	
Comments: LONGTERM TOO for Cycle 2			
(6)	C2-LONGTERM-TOO-WDSN-6-LOWRES	nearby white dwarf supernova	
Comments: LONGTERM TOO for Cycle 2			
(7)	C3-LONGTERM-TOO-WDSN-7-MEDRES	nearby white dwarf supernova	
Comments: LONGTERM TOO for Cycle 3			
(8)	C3-LONGTERM-TOO-WDSN-8-LOWRES	nearby white dwarf supernova	
Comments: LONGTERM TOO for Cycle 3			
(9)	C3-LONGTERM-TOO-WDSN-9-LOWRES	nearby white dwarf supernova	
Comments: LONGTERM TOO for Cycle 3			

Observation

Proposal 2072, Observation 1: LRS-Slit

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(LRS-Slit (Obs 1)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	SN2021AEFX	RA: 04 19 53.4000 (64.9725000d) Dec: -54 56 53.09 (-54.94808d) Equinox: J2000		

Comments:
Category=Star
Description=[Type Ia supernovae]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FASTGRPAVG8	4	1	1	88.801	114476

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	4	1	1	1	1	11.1		F1000W

Proposal 2072 - Observation 1 - See Through Supernovae: Nebular Spectroscopy of Exploding White Dwarfs

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	134	2	4	1	2	1492.972	114476

Special Requirements

Before Date 01-SEP-2022:00:00:00

Group Observations 1, 2, Non-interruptible

Observation

Proposal 2072, Observation 2: NIR PRISM

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	SN2021AEFX	RA: 04 19 53.4000 (64.9725000d) Dec: -54 56 53.09 (-54.94808d) Equinox: J2000		

Comments:
Category=Star
Description=[Type Ia supernovae]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	114476

Template

Slit	Subarray
S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	2	4	1	NONE	3	12	525.2	114476

Special Requirements	Group Observations 1, 2, Non-interruptible

Observation

Proposal 2072, Observation 3: LRS-Slit

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(LRS-Slit (Obs 3)) Warning (Form): Groups/Int cannot be 1, Groups/Int = 2 requires permission and Groups/Int of 3-4 is allowed but not recommended.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	SN2021AEFX	RA: 04 19 53.4000 (64.9725000d) Dec: -54 56 53.09 (-54.94808d) Equinox: J2000		

Comments:
Category=Star
Description=[Type Ia supernovae]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FASTGRPAVG8	4	1	1	88.801	114476

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	4	1	1	1	1	11.1		F1000W

Proposal 2072 - Observation 3 - See Through Supernovae: Nebular Spectroscopy of Exploding White Dwarfs

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	134	2	4	1	2	1492.972	114476

Special Requirements

Between Dates 30-OCT-2022:00:00:00 and 08-JAN-2023:00:00:00

Group Observations 3, 4, Non-interruptible

Observation

Proposal 2072, Observation 4: NIR PRISM

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	SN2021AEFX	RA: 04 19 53.4000 (64.9725000d) Dec: -54 56 53.09 (-54.94808d) Equinox: J2000		

Comments:
Category=Star
Description=[Type Ia supernovae]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	114476

Template

Slit	Subarray
S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	5	2	1	NONE	3	6	525.2	114476

Special Requirements	
	Group Observations 3, 4, Non-interruptible

Observation

Proposal 2072, Observation 5: LRS-Slit

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(12)	SN2022PUL	RA: 12 26 48.8460 (186.7035250d) Dec: +08 26 55.32 (8.44870d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	6	1	1	16.65	156132

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	6	1	1	1	1	16.65		F1000W

Proposal 2072 - Observation 5 - See Through Supernovae: Nebular Spectroscopy of Exploding White Dwarfs

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	89	3	6	1	2	1492.972	156132

Special Requirements

Between Dates 01-MAY-2023:00:00:01 and 01-AUG-2023:00:00:01

7 After 5 by 174 Days to 251 Days
Group Observations 5, 6, Non-interruptible

Observation

Proposal 2072, Observation 6: NIR PRISM

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(12)	SN2022PUL	RA: 12 26 48.8460 (186.7035250d) Dec: +08 26 55.32 (8.44870d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	156132

Template

Slit	Subarray
S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	2	4	1	NONE	3	12	525.2	156132

Special Requirements	Group Observations 5, 6, Non-interruptible

Observation

Proposal 2072, Observation 7: LRS-Slit

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 7:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(12)	SN2022PUL	RA: 12 26 48.8460 (186.7035250d) Dec: +08 26 55.32 (8.44870d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FASTGRPAVG8	4	1	1	88.801	156132

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	32	1	1	1	1	88.801		F1000W

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	89	5	10	1	2	2491.986	156132

Special Requirements

7 After 5 by 174 Days to 251 Days
Group Observations 7, 8, Non-interruptible

Observation

Proposal 2072, Observation 8: NIR PRISM

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(12)	SN2022PUL	RA: 12 26 48.8460 (186.7035250d) Dec: +08 26 55.32 (8.44870d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	156132

Template

Slit	Subarray
S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	10	2	1	NONE	3	6	962.867	156132

Special Requirements	Group Observations 7, 8, Non-interruptible

Observation

Proposal 2072, Observation 9: MRS

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(11)	SN2022AAIQ	RA: 14 26 32.0290 (216.6334542d) Dec: +56 35 3.13 (56.58420d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FASTGRPAVG8	4	1	1	88.801	147857

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	50	4	1	Dither 1	4	16	2253.332	149265
1	SHORT(A)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
1	SHORT(A)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
2		IMAGER	F770W	FASTR1	50	4	1	Dither 1	4	16	2253.332	149265
2	MEDIUM(B)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
2	MEDIUM(B)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
3		IMAGER	F1000W	FASTR1	50	4	1	Dither 1	4	16	2253.332	149265
3	LONG(C)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
3	LONG(C)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265

Special Requirements	
	After Date 12-MAR-2023:00:00:00 11 After 9 by 80 Days to 150 Days Group Observations 9, 10, Non-interruptible

Observation

Proposal 2072, Observation 10: NIR G395M+G235M

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(11)	SN2022AAIQ	RA: 14 26 32.0290 (216.6334542d) Dec: +56 35 3.13 (56.58420d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	147857

Template

Slit	Subarray
S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395M/F290LP	S200A1	NRSIRS2RAPID	19	1	1	NONE	3	3	875.333	147857
2	G235M/F170LP	S200A1	NRSIRS2RAPID	19	1	2	NONE	3	3	875.333	147857

Special Requirements	Group Observations 9, 10, Non-interruptible

Observation

Proposal 2072, Observation 11: MRS

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(11)	SN2022AAIQ	RA: 14 26 32.0290 (216.6334542d) Dec: +56 35 3.13 (56.58420d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FASTGRPAVG8	4	1	1	88.801	147857

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	50	4	1	Dither 1	4	16	2253.332	149265
1	SHORT(A)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
1	SHORT(A)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
2		IMAGER	F770W	FASTR1	50	4	1	Dither 1	4	16	2253.332	149265
2	MEDIUM(B)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
2	MEDIUM(B)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
3		IMAGER	F1000W	FASTR1	50	4	1	Dither 1	4	16	2253.332	149265
3	LONG(C)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265
3	LONG(C)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	149265

Special Requirements	
	11 After 9 by 80 Days to 150 Days Group Observations 11, 12, Non-interruptible

Observation

Proposal 2072, Observation 12: NIR G395M+G235M

Thu May 18 03:00:37 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(11)	SN2022AAIQ	RA: 14 26 32.0290 (216.6334542d) Dec: +56 35 3.13 (56.58420d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae, Type Ia supernovae]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	147857

Template

Slit	Subarray
S200A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395M/F290LP	S200A1	NRSIRS2RAPID	19	1	1	NONE	3	3	875.333	147857
2	G235M/F170LP	S200A1	NRSIRS2RAPID	19	1	2	NONE	3	3	875.333	147857

Special Requirements	
	Group Observations 11, 12, Non-interruptible